CO2-AT1-PYTHON SQLLITE

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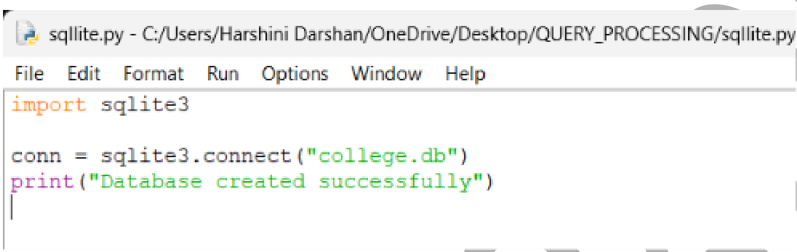
Activity 1: Create Database

import sqlite3

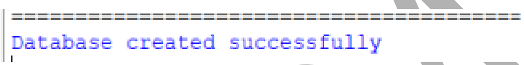
conn = sqlite3.connect("college.db")

print("Database created successfully")




```
import sqlite3

conn = sqlite3.connect("college.db")
print("Database created successfully")
```

Output



```
========================================
Database created successfully
```

Activity 2: Create Table

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE Student(

id INTEGER PRIMARY KEY,

name TEXT,

age INTEGER,

department TEXT

)

""")

conn.commit()

print("Table Created Successfully")

File Edit Format Run Options Window Help

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE Student(
id INTEGER PRIMARY KEY,
name TEXT,
age INTEGER,
department TEXT
)
""")

conn.commit()

print("Table Created Successfully")
```

Output

```
Table Created Successfully
```

Run this only once. Running it again gives:

sqlite3.OperationalError: table Student already exists

Activity 3: Insert One Record

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("""

INSERT INTO Student

VALUES(1,'Ravi',20,'CSE')

""")

conn.commit()

print("1 Record Inserted")

File Edit Format Run Options Window Help

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
INSERT INTO Student
VALUES(1,'Ravi',20,'CSE')
""")

conn.commit()

print("1 Record Inserted")
```

Output

```
==========================
1 Record Inserted
```

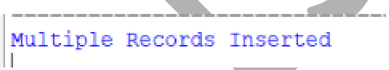
```
================================
(1, 'Ravi', 20, 'CSE')
```

Activity 4: Insert Multiple Records

import sqlite3

```
conn = sqlite3.connect("college.db")
cursor = conn.cursor()

students = [
(2,'Priya',21,'IT'),
(3,'Amit',19,'ECE'),
(4,'Neha',22,'CSE')
]

cursor.executemany(
"INSERT INTO Student VALUES(?,?,?,?)",
students
)

conn.commit()

print("Multiple Records Inserted")
```

Output



Activity 5: Display All Records

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()
```

cursor.execute("SELECT * FROM Student")

rows = cursor.fetchall()

for row in rows:

   print(row)

```
File   Edit   Format   Run   Options   Window   Help
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("SELECT * FROM Student")

rows = cursor.fetchall()

for row in rows:
    print(row)
```

Output

```
==================================
(1, 'Ravi', 20, 'CSE')
(2, 'Priya', 21, 'IT')
(3, 'Amit', 19, 'ECE')
(4, 'Neha', 22, 'CSE')
```

Activity 6: Select Specific Columns

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("SELECT name,age FROM Student") rows = cursor.fetchall()

for row in rows:

    print(row)

```
sqllite.py - C:/Users/Harshini Darshan/OneDrive/Desktop/QUERY_PROCE
File  Edit  Format  Run  Options  Window  Help
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("SELECT name,age FROM Student")

rows = cursor.fetchall()

for row in rows:
    print(row)
```

Output

```
============================
('Ravi', 20)
('Priya', 21)
('Amit', 19)
('Neha', 22)
```

Activity 7: WHERE Clause

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("SELECT * FROM Student WHERE age>20")

rows = cursor.fetchall()

for row in rows:

  print(row)

```
File   Edit   Format   Run   Options   Window   Help
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("SELECT * FROM Student WHERE age>20")

rows = cursor.fetchall()

for row in rows:
    print(row)
```

Output

```
==================================
(2, 'Priya', 21, 'IT')
(4, 'Neha', 22, 'CSE')
```

Activity 8: ORDER BY

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("SELECT * FROM Student ORDER BY age")

rows = cursor.fetchall()

for row in rows:

  print(row)

File Edit Format Run Options Window Help

```python
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("SELECT * FROM Student ORDER BY age")

rows = cursor.fetchall()

for row in rows:
    print(row)
```

Output

```
======================================
(3, 'Amit', 19, 'ECE')
(1, 'Ravi', 20, 'CSE')
(2, 'Priya', 21, 'IT')
(4, 'Neha', 22, 'CSE')
```

Activity 9: Update Record

```python
import sqlite3


conn = sqlite3.connect("college.db")

cursor = conn.cursor()


cursor.execute("""

UPDATE Student

SET age=23

WHERE id=4

""")


conn.commit()
```

print("Record Updated")

```
File   Edit   Format   Run   Options   Window   Help
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
UPDATE Student
SET age=23
WHERE id=4
""")

conn.commit()

print("Record Updated")
```

Output

```
================================
Record Updated
```

Activity 10: View Updated Table

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("SELECT * FROM Student")

rows = cursor.fetchall()

for row in rows:

    print(row)

File Edit Format Run Options Window Help

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("SELECT * FROM Student")

rows = cursor.fetchall()

for row in rows:
    print(row)
```

Output

```
====================================
(1, 'Ravi', 20, 'CSE')
(2, 'Priya', 21, 'IT')
(3, 'Amit', 19, 'ECE')
(4, 'Neha', 23, 'CSE')
```

Activity 11: Delete Record

import sqlite3

conn = sqlite3.connect("college.db")

cursor = conn.cursor()

cursor.execute("""

DELETE FROM Student

WHERE id=3

""")

conn.commit()

print("Record Deleted")

File Edit Format Run Options Window Help

```python
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
DELETE FROM Student
WHERE id=3
""")

conn.commit()

print("Record Deleted")
```

Output

```
======================
Record Deleted
```

Activity 12: View Final Table

```python
import sqlite3


conn = sqlite3.connect("college.db")

cursor = conn.cursor()


cursor.execute("SELECT * FROM Student")


rows = cursor.fetchall()


for row in rows:
    print(row)
```

File Edit Format Run Options Window Help

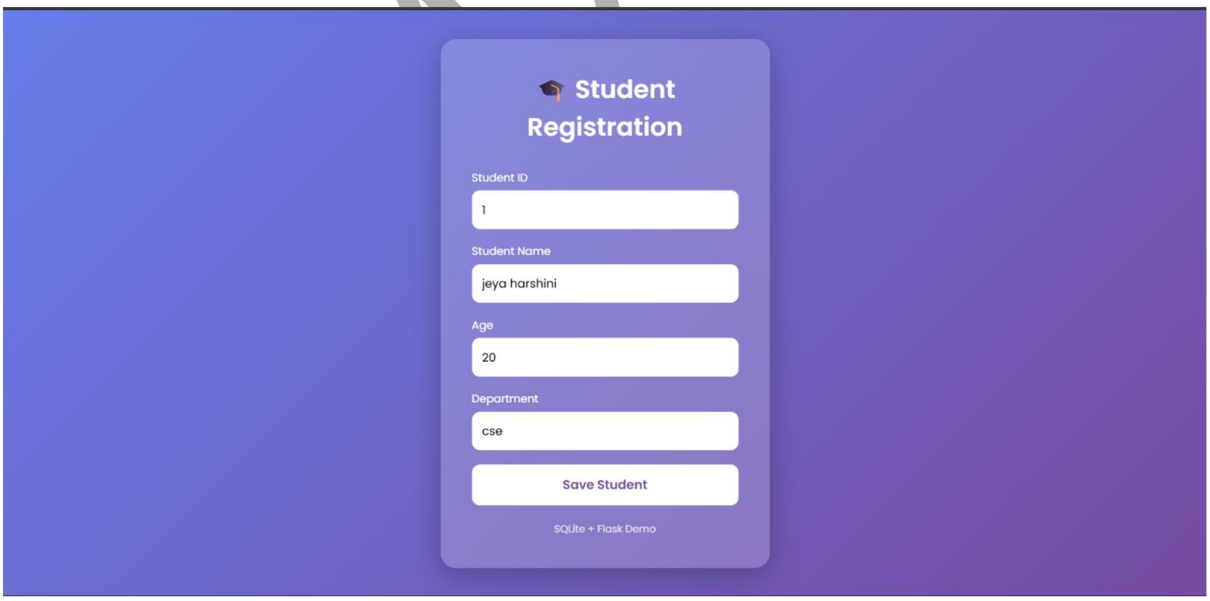
```python
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("SELECT * FROM Student")

rows = cursor.fetchall()

for row in rows:
    print(row)
```

Output

```
================================
(1, 'Ravi', 20, 'CSE')
(2, 'Priya', 21, 'IT')
(4, 'Neha', 23, 'CSE')
```


APPLICATION:-

Student registration



Backend database

```
PS C:\Users\Harshini Darshan\OneDrive\Desktop\StudentApp> python view.py
(1, 'jeya harshini', 20, 'cse')
(2, 'subhaashree', 18, 'aids')
```